

Multiobjective Optimization

Interactive Methods: Practical Examples

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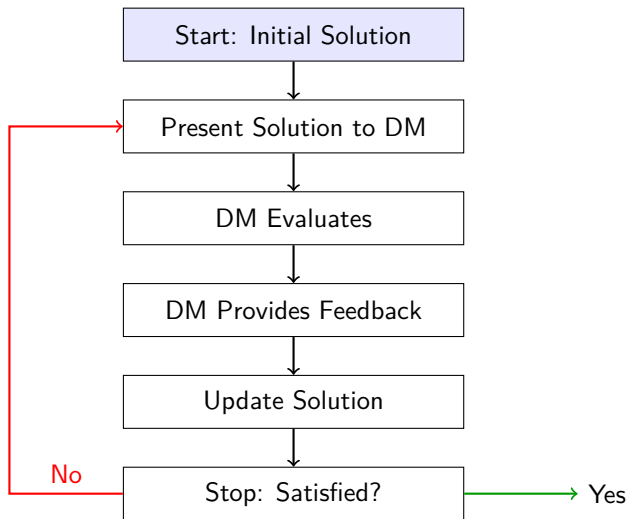
Key Concepts:

- Preferences articulated **during** solution process
- Bidirectional information flow
- Methods covered:
 - 1 Step Method (Benayoun et al., 1971)
 - 2 Geoffrion-Dyer-Feinberg (1972)
 - 3 Interactive Goal Programming (Dyer, 1972)
 - 4 Zionts-Wallenius (1976)

Today:

- **Practical examples** of interactive methods
- Real-world applications
- Lessons learned from implementations

General Procedure



Problem: Water quality management in a river basin

Objectives:

- ① Z_1 : Minimize treatment costs
- ② Z_2 : Maximize water quality (DO level)
- ③ Z_3 : Maximize equity among dischargers

Decision Makers:

- Environmental Protection Agency (EPA)
- State environmental agencies
- Industrial representatives
- Municipalities

Payoff Table Construction

Solution	Z_1 (Cost)	Z_2 (DO)	Z_3 (Equity)
$\mathbf{x}^1$ (min cost)	100	4.0	0.30
$\mathbf{x}^2$ (max DO)	200	8.5	0.45
$\mathbf{x}^3$ (max equity)	250	6.0	0.95
Min	100	4.0	0.30
Max	250	8.5	0.95

Note: Z_1 is to be minimized (convert to maximization)

For each objective:

$$\alpha_k = \frac{M_k - n_k}{\left(\sum_{j=1}^n (c_j^k)^2\right)^{1/2}}$$

Assume normalized coefficients:

Objective	$M_k - n_k$	α_k	π_k
Z_1 (cost)	150	0.30	0.25
Z_2 (DO)	4.5	0.45	0.38
Z_3 (equity)	0.65	0.45	0.37

Result: $\pi = (0.25, 0.38, 0.37)$

Problem:

$$\begin{aligned} & \text{minimize } \lambda \\ & \text{s.t. } 0.25(M_1 - Z_1) \leq \lambda \\ & \quad 0.38(M_2 - Z_2) \leq \lambda \\ & \quad 0.37(M_3 - Z_3) \leq \lambda \\ & \quad \mathbf{x} \in \mathbf{F}_d \end{aligned}$$

Solution:

$$Z_1 = 180, \quad Z_2 = 6.5, \quad Z_3 = 0.55$$

DM Evaluation: "Cost is too high, DO is acceptable, equity is acceptable"

Second Iteration

DM Response: Relax objective Z_2 (DO) by $\Delta Z_2 = 1.0$

New Constraint:

$$Z_2 \geq 6.5 - 1.0 = 5.5$$

New Problem:

$$\begin{aligned} & \text{minimize } \lambda \\ \text{s.t. } & 0.25(M_1 - Z_1) \leq \lambda \\ & 0.37(M_3 - Z_3) \leq \lambda \\ & Z_2 \geq 5.5 \\ & \mathbf{x} \in \mathbf{F}_d \end{aligned}$$

Solution:

$$Z_1 = 150, \quad Z_2 = 5.5, \quad Z_3 = 0.60$$

Third Iteration

DM Evaluation: "Cost is now acceptable, equity could be improved"

DM Response: Relax objective Z_1 (cost) by $\Delta Z_1 = 20$

New Constraint:

$$Z_1 \leq 150 + 20 = 170$$

New Problem:

$$\begin{aligned} & \text{minimize } \lambda \\ & \text{s.t. } 0.37(M_3 - Z_3) \leq \lambda \\ & \quad Z_1 \leq 170, \quad Z_2 \geq 5.5 \\ & \quad \mathbf{x} \in \mathbf{F}_d \end{aligned}$$

Solution:

$$Z_1 = 170, \quad Z_2 = 5.5, \quad Z_3 = 0.75$$

DM Evaluation: Satisfied!

Facility Location Application

Problem: Location of energy facilities

Objectives:

- 1 Z_1 : Minimize cost
- 2 Z_2 : Minimize environmental impact
- 3 Z_3 : Maximize equity

Decision Maker: State energy planning board

Challenge: Decision makers prefer to evaluate alternatives visually

DM Asked:

"At the current solution, how much Z_2 are you willing to sacrifice to gain one unit of Z_1 (relative to reference Z_3)?"

DM Response: $MRS_{12} = 0.5$, $MRS_{32} = 0.8$

Direction Problem:

$$\text{maximize } 0.5\nabla Z_1 + 1.0\nabla Z_2 + 0.8\nabla Z_3$$

$$\text{s.t. } \mathbf{y} \in \mathbf{F}_d$$

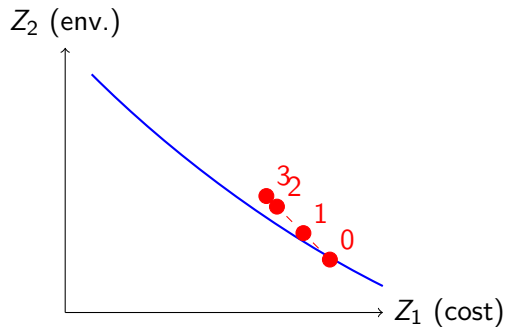
Alternatives along direction:

t	Z_1 (cost)	Z_2 (env.)	Z_3 (equity)	DM Choice
0.00	200	0.50	0.40	✓
0.25	180	0.45	0.45	
0.50	160	0.40	0.50	
0.75	140	0.35	0.55	
1.00	120	0.30	0.60	

DM chooses $t = 0.50$

- Balanced solution
- Tradeoffs seem reasonable

Iteration History



Convergence: 4 iterations to best-compromise

Problem: Allocate budget across departments

Objectives:

- ① Z_1 : Maximize student enrollment
- ② Z_2 : Maximize research output
- ③ Z_3 : Maximize faculty satisfaction

Decision Maker: University Dean

Initial Goals:

$$G_1 = 1000, \quad G_2 = 200, \quad G_3 = 0.80$$

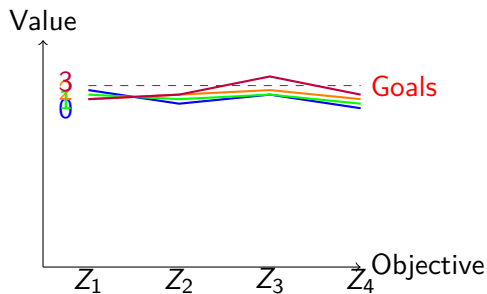
Goal Adjustments

Iteration	G_1	G_2	G_3	Solution
0	1000	200	0.80	$Z = (980, 180, 0.75)$
1	950	200	0.80	$Z = (950, 185, 0.76)$
2	950	190	0.82	$Z = (930, 190, 0.80)$
3	920	190	0.85	$Z = (920, 188, 0.84)$

DM Adjustments:

- Iteration 1: Reduce enrollment target (too ambitious)
- Iteration 2: Increase research and satisfaction targets
- Iteration 3: Satisfied with tradeoffs

Value Path Display



Progress: Values approach goals over iterations

Multiobjective Investment Planning

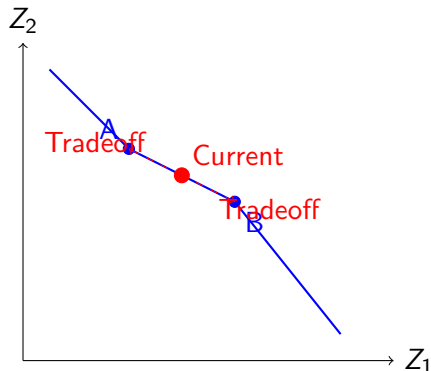
Problem: Select portfolio of projects

Objectives:

- 1 Z_1 : Maximize net present value
- 2 Z_2 : Maximize employment
- 3 Z_3 : Minimize environmental impact

Procedure:

- 1 Start with noninferior solution
- 2 Identify adjacent noninferior solutions
- 3 Ask DM: "Do you prefer any of these?"
- 4 Move to preferred solution
- 5 Repeat until no preference



DM Choice: "I prefer B over current" $\rightarrow$ move to B

At each iteration:

"Consider moving from the current solution to alternative A. This would:

- *Increase Z_1 by 10 units*
- *Decrease Z_2 by 5 units*
- *Increase Z_3 by 2 units*

Do you prefer this tradeoff?"

DM Response: Yes/No/Indifferent

If Yes: Move to alternative **If No:** Try another alternative **If Indifferent:** Stop (best-compromise found)

Performance of Interactive Methods

Method	Iterations	DM Effort	Solution Quality
Step Method	3-5	Low	Good
Geoffrion	3-8	High	Excellent
Interactive GP	3-6	Medium	Good
Zionts-Wallenius	4-10	Medium	Good

Key Insights:

- More DM effort → better solution quality
- Step Method: Easiest for DM
- Geoffrion: Best solution but more demanding

Success Factors:

① Clear Communication

- Use visual displays (value paths, graphs)
- Explain tradeoffs in plain language

② Gradual Learning Process

- Decision makers discover preferences
- Tradeoffs become clearer over iterations

③ Flexible Interaction

- Allow DM to explore alternatives
- Let DM backtrack if needed

Pitfall: "Too much information can overwhelm the decision maker"

Experiment Summary:

- Compared Step Method, Geoffrion, and trial-and-error
- Subjects: Business students and managers
- Multiobjective decision problem

Results:

- 1 **Ease of Use:** Trial-and-error $\succ$ Step $\succ$ Geoffrion
- 2 **Confidence:** Trial-and-error $\approx$ Geoffrion $\succ$ Step
- 3 **Overall Preference:** Trial-and-error $\succ$ Geoffrion $\approx$ Step

Implication:

"Simple exploration may be more natural than formal interactive procedures."

When to Use Interactive Methods:

- 1 Decision maker is available and willing
- 2 Preferences are not well-defined initially
- 3 Learning about tradeoffs is important
- 4 Multiple stakeholders need to be involved

Practical Tips:

- Start with generating methods to show range
- Use visual displays (value paths, graphs)
- Allow DM to explore freely
- Document preferences and reasoning
- Conduct sensitivity analysis

Key Takeaways

Interactive Methods - Practical Examples:

- **Step Method:** Water quality management (3 iterations)
- **Geoffrion Method:** Energy facility location (4 iterations)
- **Interactive GP:** Academic resource allocation (3 iterations)
- **Zionts-Wallenius:** Investment planning (adjacent solutions)

Key Lessons:

- 1 Interaction helps DMs discover preferences
- 2 Visual displays are essential
- 3 Formal methods may not beat simple exploration
- 4 DMs prefer to learn through tradeoffs

Next: Multiple-Decision-Maker Methods

Topics:

- ① Aggregation of Individual Preferences
 - Welfare economics
 - Social choice theory
 - Arrow's impossibility theorem
- ② Counseling Individual Decision Makers
- ③ Predicting Political Outcomes
 - Paretian analysis
 - Game theory
 - Vote-trading models

Key Question: How do we handle multiple conflicting decision makers?



COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.